Towards a Unified Copernicus Foundation Model for Earth Vision

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Abstract

Advances in Earth observation (EO) foundation models have unlocked the potential of big satellite data to learn generic representations from space, benefiting a wide range of downstream applications crucial to our planet. However, most existing efforts remain limited to fixed spectral sensors, focus solely on the Earth's surface, and overlook valuable metadata beyond imagery. In this work, we take a step towards next-generation EO foundation models with three key components: 1) Copernicus-Pretrain, a massive-scale pretraining dataset that integrates 18.7M aligned images from all major Copernicus Sentinel missions, spanning from the Earth's surface to its atmosphere; 2) Copernicus-FM, a unified foundation model capable of processing any spectral or non-spectral sensor modality using extended dynamic hypernetworks and flexible metadata encoding; and 3) Copernicus-Bench, a systematic evaluation benchmark with 15 hierarchical downstream tasks ranging from preprocessing to specialized applications for each Sentinel mission. Our dataset, model, and benchmark greatly improve the scalability, versatility, and multimodal adaptability of EO foundation models, while also creating new opportunities to connect EO, weather, and climate research. Codes at https://github.com/zhu-xlab/Copernicus-FM.

1. Introduction

Earth observation (EO) satellites provide a critical means of monitoring planetary dynamics, capturing land cover changes, atmospheric composition, and other important environmental phenomena [47, 64, 67]. Recent advances in self-supervised learning have revolutionized the usage of big EO data, enabling the development of general-purpose foundation models to learn generalized representations from unlabeled imagery at scale [54, 69]. Despite rapid progress, the evolution of EO foundation models faces critical limitations in three dimensions: sensor diversity, model flexibility, and evaluation breadth.

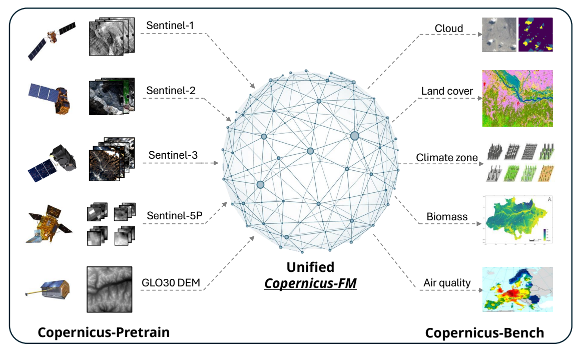

Figure 1. Overview of our efforts towards a unified Copernicus foundation model, from pretraining to benchmarking.

First, current pretraining datasets predominantly focus on high- to medium-resolution sensors like Sentinel-1/2 [17, 52] and Landsat [60] observing the Earth's surface [6, 49, 56]. This excludes lower-resolution but temporally rich missions like Sentinel-3 [16] and Sentinel-5P [53], which provide near-daily global coverage of land, oceanic, and atmospheric variables critical for climate studies. Second, most existing EO foundation models adopt rigid architectures tailored to one or more specific sensor modalities, lacking the capacity to dynamically adapt to new spectral bands or non-spectral input. While recent efforts [10, 62] introduce some spectral flexibility, they still lack mechanisms to handle non-spectral variables that possess a significant amount of EO data. Furthermore, current foundation model evaluation benchmarks primarily focus on surface applications using RGB, multispectral, or SAR sensors [34, 40], while overlooking coarse-scale sensors and atmospheric tasks. These limitations hinder the development of versatile multimodal foundation models that integrate EO with weather and climate research, highlighting both a critical challenge and a promising opportunity in the era of global environmental change.

To address these challenges, we introduce three synergistic contributions that enhance the scalability, versatility, and multimodal integration of EO foundation models. First, we introduce **Copernicus-Pretrain**, one of the largest and most diverse EO pretraining datasets to date, comprising 18.7 million aligned observations from all major operational Copernicus Sentinel missions (Sentinel-1 to Sentinel-5P). Unlike prior datasets focusing on fine-grained surface observations, Copernicus-Pretrain enables holistic modeling of Earth system interactions by integrating atmospheric variables and coarse-scale observations with wider and more frequent coverage. Second, we propose **Copernicus-FM**, a unified foundation model capable of processing any spectral or non-spectral sensor using dynamic hypernetworks. With additional support for metadata integration, it is valuable for a wide range of practical applications. Third, we establish **Copernicus-Bench**, a comprehensive evaluation benchmark with 15 downstream tasks hierarchically organized across preprocessing (e.g., cloud removal), base applications (e.g., land cover classification), and specialized applications (e.g., air quality estimation). This benchmark enables systematic assessment of foundation model performances across various Sentinel missions on different levels of practical applications.

Our work opens new frontiers in three directions: (1) scaling EO pretraining to unified multimodal datasets, breaking the traditional silos between surface and atmospheric observations; (2) verifying that dynamic architectures can overcome sensor heterogeneity, a longstanding challenge in multi-source remote sensing; and (3) providing the first evidence that joint cross-modal pretraining enhances performance on both surface and atmospheric tasks. Furthermore, our efforts create novel opportunities to integrate EO foundation models with weather and climate prediction systems—for instance, using compressed EO embeddings (with rich semantic information) in addition to geographical coordinates to support climate modeling.

2. Related work

EO pretraining datasets Early efforts like fMoW [13], Million-AID [37], and SEN12MS [48] pioneered large-scale pretraining with supervised datasets before the era of self-supervised learning. SeCo [39] introduced Gaussian sampling around populated regions to enrich the landscape diversity with multiseasonal time series, which was further extended by SSL4EO-S12 [56] and SSL4EO-L [49] through overlap and NaN filtering on Sentinel-1/2 and Landsat series. SatlasPretrain [6] and Major TOM [21] boost the dataset sizes through more dense global coverage. Recently, increasing efforts have been spent to broaden sensor diversity such as SpectralEarth [12] for hyperspectral pretraining and MMEarth [42] that gathers Sentinel-1/2, elevation, and several EO products together. Our Copernicus-Pretrain dataset aligns all primary Sentinel missions (1–5P) with extended global coverage (e.g., polar regions), enabling joint surface–atmosphere modeling at scale.

EO foundation models Single-sensor models dominate early foundation model research, which can be categorized based on pretraining strategies into: 1) contrastive methods like GASSL [5], SeCo [39], MATTER [1], CACo [38], SatMIP [11], etc., 2) masked image modeling (MIM) methods such as SatMAE [15], Prithvi [32], SpectralGPT [30], Scale-MAE [46], and many others [36, 42, 43, 50, 51, 59], and 3) hybrid methods like GFM [41], SoftCon [58], SAR-JEPA [35], etc. Most of these models focus on optical data. In a trend towards multimodal pretraining, mainstream approaches use either separate encoders (DeCUR [57], CROMA [22], SkySense [24], etc.) or joint encoders with a few fixed modalities [2, 3, 25, 31, 55, 63, 66]. While flexible multimodal architectures like FoMo-Net [10], SenPa-MAE [45], and DOFA [62] support input with any channels, they are restricted to spectral sensors and ignore metadata. Our Copernicus-FM enhances this approach through dynamic hypernetworks that adapt to arbitrary spectral/non-spectral inputs and support metadata integration, effectively unifying surface and atmospheric data streams.

EO benchmarks Existing benchmarks for evaluating EO foundation models vary widely in scope and focus but are, in general, limited in sensor and task diversity. SustainBench [65] targets sustainable development goals with 15 socioeconomic tasks, GEO-Bench [34] standardizes 12 classification/segmentation tasks with a majority focusing on optical imagery, PhilEO Bench [20] consists of three Sentinel-2-derived tasks, and FoMo-Bench [10] focuses on forest monitoring with 15 multimodal datasets. Recent efforts like PANGAEA [40] address geographical bias and model generalizability by aggregating diverse datasets but remain surface-centric. Our Copernicus-Bench fills the gap with 15 hierarchical tasks spanning three application levels, covering all major Sentinel missions and encompassing both surface and atmosphere.

3. Copernicus-Pretrain

We introduce a unified dataset, Copernicus-Pretrain, containing aligned imagery organized in dense regional grids from all major Sentinel missions in operation (Sentinel-1 SAR, Sentinel-2 multispectral reflectance, Sentinel-3 multispectral radiance, and Sentinel-5P atmospheric variables), as well as an elevation product Copernicus DEM GLO-30 [19]. Copernicus-Pretrain significantly extends the scale in data size and modality range of existing EO foundation model research, and provides opportunities for bridging EO and weather/climate studies.

3.1. Data collection

The Copernicus-Pretrain dataset is designed for consistency with the ERA5 [29] reanalysis dataset to provide a conve-

Table 1. Copernicus-Pretrain dataset statistics.

	Modality	GSD	Image size	# Grid cells	# Patches	# Timestamps	# Total images
Sentinel-1 GRD	SAR	10 m	264×264×2	247,723	1,067,267	∼4	4,227,387
Sentinel-2 TOA	MS	10 m	264×264×13	247,723	1,067,267	∼4	4,218,065
Sentinel-3 OLCI	MS	300 m	96×96×21	281,375	281,375	∼8	2,189,561
Sentinel-5P CO	atmos.	1 km	28×28	306,097	306,097	1–12	2,104,735
Sentinel-5P NO2	atmos.	1 km	28×28	291,449	291,449	1–12	1,752,558
Sentinel-5P SO2	atmos.	1 km	28×28	262,259	262,259	1–12	1,366,452
Sentinel-5P O3	atmos.	1 km	28×28	306,218	306,218	1–12	2,556,631
Copernicus DEM	elevation	30 m	960×960	297,665	297,665	1	297,665
Copernicus-Pretrain				312,567	3,879,597		18,713,054

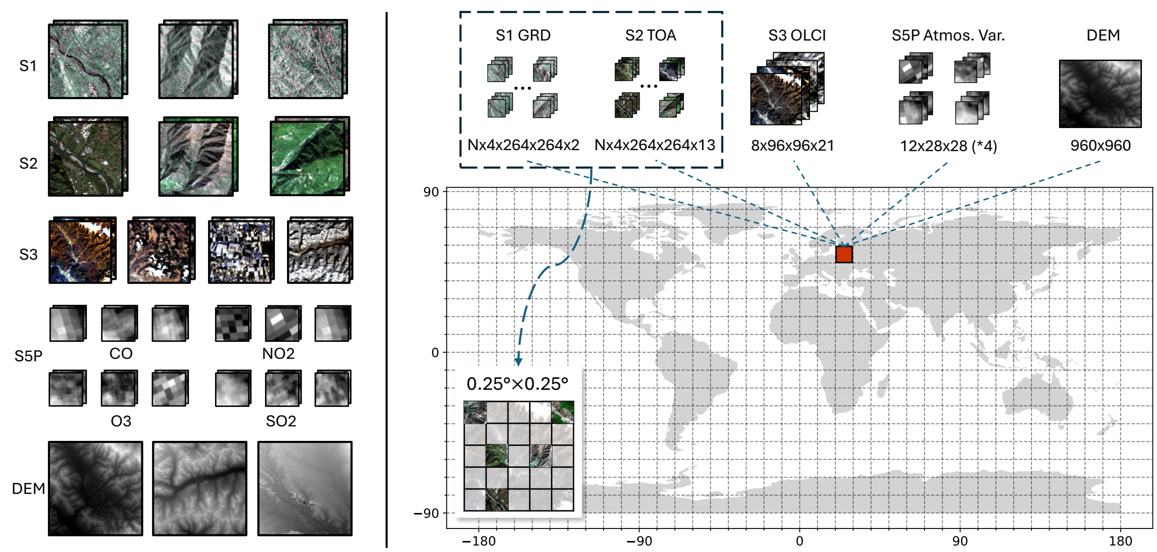

Figure 2. Schematic of the Copernicus-Pretrain dataset. N is the number of local patches. Grid cells are upscaled for ease of visualization.

nient alignment between EO imagery and weather/climate data. The workflow of dataset curation is as follows.

First, we divide the globe into ∼1M $0.25° \times 0.25°$ grid cells following the coordinate mapping of ERA5. Each cell thus covers a surface area of about 28 km × 28 km, forming the basic sample unit of the dataset. With a main interest and data availability in land and its atmosphere, we apply the land mask to filter land grids with a 0.5° buffer around the coastline, resulting in about 393K grids.

We then use Google Earth Engine [23] to download Sentinel images for each grid cell around the anchor year 2021. For Sentinel-3/5P and DEM, we cover the whole cell with patch sizes of about 96×96, 28×28, and 960×960 pixels for each corresponding modality. For Sentinel-3, we filter out cloudy tiles with bright pixel percentages above 20% and randomly download eight images from the year. For Sentinel-5P, we select four atmospheric variables: NO_2, CO, SO_2 and O_3, filter out fully NaN tiles, and download 12 monthly mean images across the year for each variable. It is worth noting that for Sentinel-3 and 5P, it is possible to have fewer images available due to cloud and NaN filtering, thus the final sequence lengths can vary for different grids. For DEM, time series is not applicable, and we simply download one image for the grid.

For Sentinel-1 and 2, the data volume is too large to cover the whole grid cell and the data can be extremely redundant. Therefore, we collect images by sampling local patches within the cells. To optimize the diversity of the data, we do not directly conduct uniform sampling inside a grid, but follow a Gaussian sampling strategy [39, 49, 56] to sample locations around top-10K populated cities with a standard deviation of 50 km. We utilize this strategy to sample ∼1M locations, each downloading 4 seasonal 264×264 image patches. We uniformly sample and download images for an additional 40K polar locations. Next, we group these local patches into the predefined $0.25° \times 0.25°$ grid cells according to their center locations. For those remaining cells that do not contain any local patches, we newly sample and download 1–2 local patches to fill in the gaps.

After the downloading process is finished, a series of quality checks are conducted, removing images that are either corrupted or still dominated by NaN values. In the end, we get ∼310K grid cells with at least one sensor available, and ∼220K grids with all modalities available. Fig. 2 (right) illustrates an example cell containing all eight modalities.

3.2. Dataset characteristics

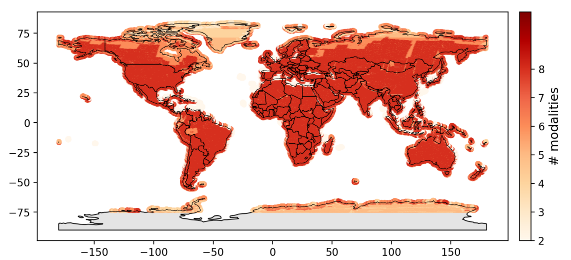

Figure 3. Global distribution of the Copernicus-Pretrain dataset.

In total, Copernicus-Pretrain consists of ∼18M images grouped into ∼310K $0.25° \times 0.25°$ cells, densely covering the whole land surface and near-land ocean with eight distinct Sentinel modalities. On average, each cell contains 4 local S1/S2 time series pairs (16 images each), 8 S3 images, 7/6/5/8 S5P $CO/NO_2/SO_2/O_3$ images, and 1 DEM image.

Fig. 3 shows the global distribution of the full dataset, while the detailed statistics are shown in Tab. 1. The distribution and statistics of the all-modality-aligned subset and other detailed analyses can be found in the appendix.

4. Copernicus-FM

Based on the Copernicus-Pretrain dataset, we introduce Copernicus-FM, a new EO foundation model that 1) can process any spectral or non-spectral input modalities with varied spatial resolutions, and 2) when available, flexibly integrates metadata information like geolocation, geographical area, and time. Fig. 4 illustrates the general pretraining framework of Copernicus-FM, which uses unified hypernetworks to dynamically patchify different modalities into patch tokens, integrates metadata as Fourier encodings added to the patch tokens, and performs masked image modeling (MIM) with auxiliary continual distillation as the training objectives.

4.1. Model architecture

Given a grid sample unit with all modalities, one image is sampled from the local patches (for S1/S2) and time series for each modality. This results in 8 images with varied spatial and channel sizes from distinct modalities, serving as the main input to the model.

Dynamic patch embedding via sensor-aware hypernetworks We use a single unified architecture to deal with different input modalities. This is implemented by extending the wavelength-conditioned dynamic patch embedding design from DOFA [62], where the core idea is to use a hypernetwork to dynamically generate kernel weights for the 2D convolution patch embedding layer.

As presented in Fig. 4, denoting one spectral image (e.g., from S1/2/3) as $\mathbf{X} \in \mathbb{R}^{C \times H \times W}$, each channel has a corresponding central wavelength $\lambda \in \{\lambda_1, ..., \lambda_C\}$ and bandwidth $\delta \in \{\delta_1, ..., \delta_C\}$ acquired from the sensor's spectral response. The wavelengths and bandwidths serve as input to the spectral hypernetwork to generate patch embedding weights.

First, we use a Fourier encoding as introduced in Bodnar et al. [8] to encode a wavelength or bandwidth value into a D-dimensional vector,

$$\mathrm{FE}(x) = \left[\cos\frac{2\pi x}{\omega_i}, \sin\frac{2\pi x}{\omega_i}\right], \quad \forall\, 0 \le i < D/2, \quad (1)$$

where ω_i are log-spaced values between the minimum and maximum:

$$\omega_i = \exp\left(\log\omega_{\min} + i \cdot \frac{\log\omega_{\max} - \log\omega_{\min}}{D/2-1}\right). \quad (2)$$

One specific Fourier encoding is designed for wavelengths and bandwidths respectively, where $\omega_{\min}$ and $\omega_{\max}$ are defined based on the corresponding value ranges. The resulting vectors $\mathbf{V}_\lambda \in \mathbb{R}^{C \times D}$ and $\mathbf{V}_\delta \in \mathbb{R}^{C \times D}$ are then added together to form the spectral encodings $\mathbf{V}_{\text{spec}} \in \mathbb{R}^{C \times D}$.

Following DOFA [62], $\mathbf{V}_{\text{spec}}$ are further transformed through MLP and multi-head attention layers to get weight vectors $\mathbf{M}_\mathbf{w} \in \mathbb{R}^{C \times p^2 D}$ and bias vectors $\mathbf{M}_\mathbf{b} \in \mathbb{R}^{C \times D}$, where p is the expected patch size. The weight vectors are then reshaped into the convolution kernel $\mathbf{K}_{\text{conv}} \in \mathbb{R}^{D \times C \times p \times p}$. Originally, a fixed patch size is required for all modalities, which can be acceptable when the scales do not differ much. However, the significant resolution gaps between our input modalities will lead to exploded memory and compute costs. To tackle this issue, we adapt the idea of FlexiVit [7] to dynamically reshape the kernel weights into a suitable patch size for each modality. Finally, the convolution operation for patch embedding is performed using the reshaped weights $\mathbf{K'}_{\text{conv}}$ and biases $\mathbf{M}_\mathbf{b}$.

The above spectral hypernetwork can process any modality with a spectral response regardless of spatial and channel dimensions. However, it can not natively process non-spectral modalities such as atmospheric constituents from S5P and elevation maps from DEM. To bridge this gap, we introduce a variable hypernetwork to generate weights for non-spectral modalities. Since these modalities do not have a common meta-attribute like wavelength

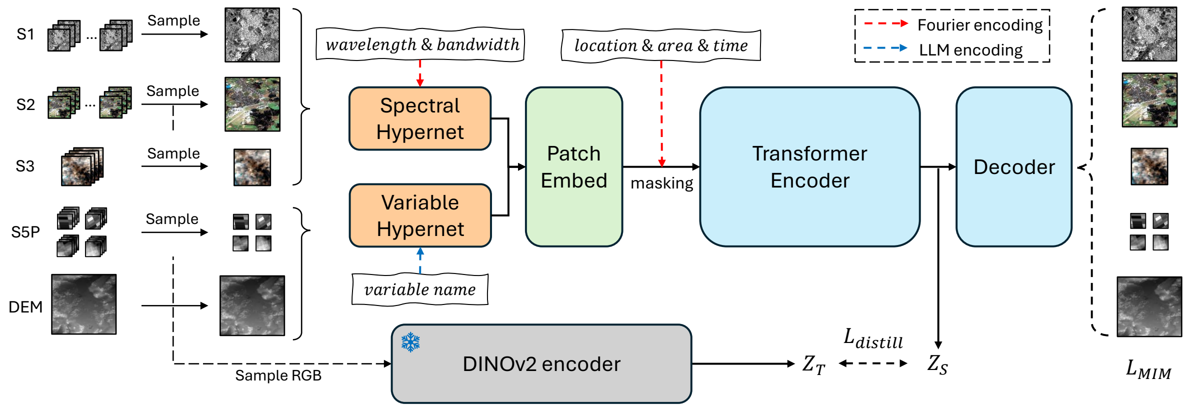

Figure 4. The general pretraining pipeline of Copernicus-FM. One image for each modality is sampled from a common grid cell in Copernicus-Pretrain, which is then patchified with kernel weights generated by the spectral or variable hypernetwork, based on the modality's spectral response or variable name. Further, Fourier-encoded metadata encodings are incorporated into the patch tokens. We conduct masked image modeling with auxiliary continual distillation for pretraining: masking and reconstructing masked-out patches for each modality, and distilling S1/2 or S2-derived RGB representations from powerful specialized teachers such as DINOv2 [44].

or bandwidth, we propose to directly encode their variable names using modern large language models (LLMs) with general scientific knowledge. Specifically, we use one frozen LLM encoder to encode the variable names of non-spectral inputs into D-dimensional vectors. This is done as a preprocessing step through one-time inference offline, thereby introducing zero additional cost to the general model framework. Similar to the spectral hypernetwork, the language-guided variable encodings $\mathbf{V}_{var}$ are further transformed through MLP and attention layers to get weight and bias vectors, and reshaped to modality-specific patch sizes for the convolution kernel of the patch embedding layer.

Metadata integration with unified Fourier encoding The dynamic patch embedding layer patchifies the input image into a sequence of patch tokens. In the standard Transformer architecture, positional encodings $\mathbf{P} \in \mathbb{R}^{N \times D}$ (where N is the number of patches) are added along with a classification token to the patch tokens. In Copernicus-FM, we introduce additional metadata encodings in parallel to the positional encodings to integrate metadata information when available. Similar to wavelength and bandwidth encoding in the spectral hypernetwork, we again use a Fourier encoding (Eqs. (1) and (2)) to unify different metadata with their corresponding value ranges.

As shown in Fig. 5, we consider three common types of metadata for one input image: geolocation, spatial coverage, and time. For geolocation, the central coordinates of one image (longitude and latitude) are encoded into $D/2$-dimensional vectors and then concatenated to the location encoding Loc $\in \mathbb{R}^D$. For spatial coverage, the patch area (in km^2) is calculated from the ground sample distance (GSD) and patch size, and then encoded into the area encoding Area $\in \mathbb{R}^D$. For time, the temporal difference (in days) between the acquisition date of the image and a reference date is calculated, and then encoded into the time encoding Time $\in \mathbb{R}^D$. These metadata encodings are further processed by a corresponding MLP layer, expanded across the patch dimensions, and added together into the positional encodings. In practice, the metadata may not always be available. Therefore, we introduce a learnable token for each metadata when it is unavailable. We simulate such metadata-missing scenarios by randomly dropping part of the metadata during each iteration.

4.2. Training objectives

Following DOFA [62], we combine MIM and continual distillation for the pretraining of Copernicus-FM. Specifically, MIM is conducted by MAE-style [27] masked reconstruction for each modality—the patch tokens are randomly masked out and sent through the encoder and a lightweight Transformer decoder to reconstruct the masked patches. A dynamic patch predictor similar to the patch embedding layer in the encoder is used for prediction. Following He et al. [27], we conduct reconstruction directly on the flattened patch tokens. Thus, a fully-connected layer is used at the end of the predictor instead of a convolutional layer.

Meanwhile, we conduct auxiliary continual distillation with a small loss weight, using powerful single-modal or general-domain foundation models to serve as an anchor to guide and refine the latent space. For example, we distill S2-derived RGB representations from a frozen DINOv2 [44] encoder with a cosine-similarity loss. This helps improve the out-of-the-box representation quality of MIM-based pretraining, reduces compute cost with faster convergence, and also implicitly boosts the model's general

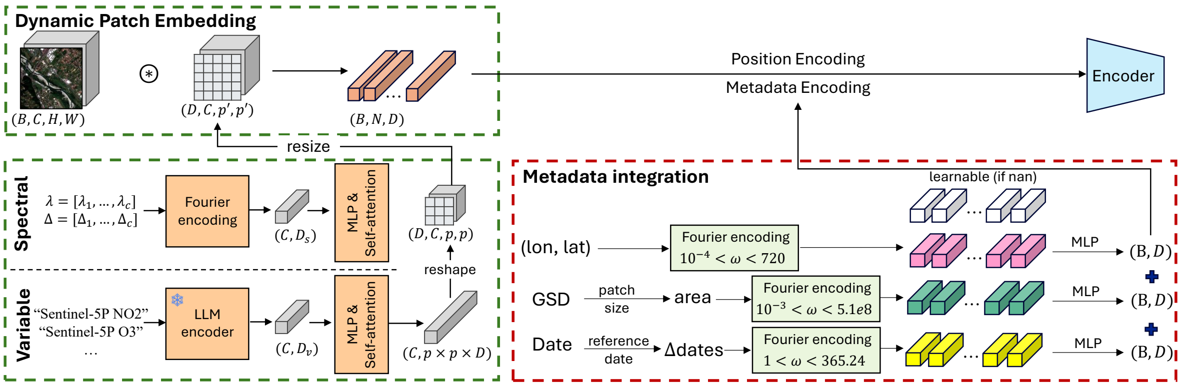

Figure 5. Dynamic patch embedding (left) and metadata integration (right) of Copernicus-FM.

Table 2. Ablation study of Copernicus-FM. OA: overall accuracy, mAP: mean average precision, and RMSE: root mean squared error.

	EuroSAT-S1	EuroSAT-S2 (OA ↑)	EuroSAT-RGB	LC100-S3 (mAP ↑)	AQ-O3-S5P (RMSE ↓)
Baseline [62] + dynamic patch size	56.3	87.6	62.2	86.7	2218.0
... + bandwidth (Fourier encoding)	56.5 ↑ 0.2	88.9 ↑ 1.3	65.4 ↑ 3.2	87.1 ↑ 0.4	1710.7 ↓ 507.3
... + variable hypernetwork	57.5 ↑ 1.0	88.9 ↑ 0.0	65.8 ↑ 0.4	86.6 ↓ 0.5	1598.1 ↓ 112.6
... + metadata encoding	77.9 ↑ **22.4**	88.9 ↑ 0.0	78.5 ↑ **12.7**	90.7 ↑ **4.1**	839.3 ↓ **758.8**
... + continual distillation	81.0 ↑ 2.9	89.5 ↑ **0.6**	78.9 ↑ 0.4	90.7 ↑ 0.0	811.6 ↓ 27.7

knowledge beyond our specific pretraining data sources.

4.3. Implementation details

We pretrain Copernicus-FM with ViT-Base on the Copernicus-Pretrain dataset (220K grids with all modalities available) for 100 epochs. Data augmentations include simple resized cropping and horizontal flipping. Input image sizes are as shown in Fig. 4, with patch sizes 16×16 for S1/2, 8×8 for S3, 4×4 for S5P, and 64×64 for DEM. We use Llama 3.2 [18] to encode variable names. The Fourier encoding value ranges are [1e2, 1e9] for wavelengths, [1, 1e9] for bandwidths, [1e-4, 720] for longitudes/latitudes, [1e-3, 5.1e8] for patch area, and [1, 365.25] for time. The drop probability for metadata is 0.7. The masking ratio for MIM is 0.7. We distill RGB from DINOv2 [44], and S1/2 from SoftCon [58] with loss weights of 0.1 and 0.2, respectively. More details can be found in the appendix.

4.4. Ablation studies

We conduct ablation studies on different components of Copernicus-FM with ViT-Small on a 10K subset of Copernicus-Pretrain, and evaluate the pretrained encoders on a range of downstream tasks covering different modalities. Specifically, we conduct k-NN classification on EuroSAT-SAR (S1) [59], EuroSAT-MS (S2) [28], and EuroSAT-RGB [28], linear probing on LC100Cls-S3 (multi-label), and dense regression with frozen encoder on AQ-O3-S5P (air pollutant regression of O_3).

The main ablation results are shown in Tab. 2, where consistent improvement can be observed when gradually adding the spectral hypernetwork bandwidth, variable hypernetwork, metadata encoding, and continual distillation. Among them, the benefits of metadata encoding appear to be the most significant, especially in non-optical modalities. This highlights the importance of metadata beyond pure imagery for remote sensing applications. Subtractive ablation, ablation on each metadata (location matters), the metadata dropping ratio (higher is better), as well as other minor ablations can be found in the appendix.

5. Copernicus-Bench

For a thorough evaluation of Copernicus-FM, we curate a benchmark suite, Copernicus-Bench, covering various Sentinel missions with hierarchical downstream tasks.

5.1. Datasets in Copernicus-Bench

Copernicus-Bench consists of 15 datasets with varied Sentinel modalities organized into three application levels in practice: preprocessing, base applications, and specialized applications. Level-1 includes two cloud detection tasks from S2/3, level-2 includes 8 land use land cover classification/segmentation tasks from S1/2/3, and level-3 includes 5 specialized tasks from S1/2/3/5P. Among all the datasets, 9 are derived from existing datasets with permissive licenses, and 6 are newly curated to fill in the gap in ML-ready

Table 3. Characteristics of datasets in Copernicus-Bench. seg, cls, cd, and reg represent segmentation, classification, change detection, and regression, respectively. *Time series support (default mode is 1 image for seg and 2 for cd). †Geolocation metadata not available.

Level	Name	Task	# Images	Image Size	# Classes	Source	License
L1	Cloud-S2	seg	1699/567/551	512×512×13	4	CloudSEN12+ [4]	CC0-1.0
	Cloud-S3	seg	1197/399/399	256×256×21	5	new	CC-BY-4.0
L2	EuroSAT-S1	cls	16200/5400/5400	64×64×2	10	EuroSAT-SAR [59]	CC-BY-4.0
	EuroSAT-S2	cls	16200/5400/5400	64×64×13	10	EuroSAT [28]	MIT
	BigEarthNet-S1	cls	11894/6117/5991	120×120×2	19	BigEarthNet v2.0 [14]	CDLA-Permissive-1.0
	BigEarthNet-S2	cls	11894/6117/5991	120×120×12	19	BigEarthNet v2.0 [14]	CDLA-Permissive-1.0
	LC100Cls-S3	cls	5181/1727/1727*	96×96×21	23	new	CC-BY-4.0
	DFC2020-S1†	seg	3156/986/986	256×256×2	10	DFC2020 [26]	CC-BY-4.0
	DFC2020-S2†	seg	3156/986/986	256×256×13	10	DFC2020 [26]	CC-BY-4.0
	LC100Seg-S3	seg	5181/1727/1727*	96×96×21 (288×288)	23	new	CC-BY-4.0
L3	Flood-S1	cd	3000/1000/1000*	224×224×2	3	Kuro Siwo [9]	MIT
	LCZ-S2†	cls	15000/5000/5000	32×32×10	17	So2Sat LCZ42 [68]	CC-BY-4.0
	Biomass-S3	reg	3000/1000/1000*	96×96×21 (288×288)	-	new	CC-BY-4.0
	AQ-NO2-S5P	reg	1480/493/494*	56×56×1	-	new	CC-BY-4.0
	AQ-O3-S5P	reg	1480/493/494*	56×56×1	-	new	CC-BY-4.0

Table 4. Benchmark results with representative single-, dual-, and multi-modal foundation models on Copernicus-Bench. We report three-run averages with standard deviations. *: patch size 16 for S1/2 (8 for S3, 4 for S5P). Random: encoder random init. Best scores in **bold**.

	Metric	Supervised	Supervised	Random	SoftCon [58]	CROMA [22]	DOFA [62]	Copernicus-FM
Backbone	–	ViT-S/16	ViT-B/16	ViT-B/16	ViT-B/14	ViT-B/8	ViT-B/16	ViT-B/16*
Modality	–	–	–	–	S1/S2	S1+S2	All (spectral)	All
Cloud-S2	mIoU	64.2 ± 0.9	59.4 ± 1.0	60.4 ± 0.2	66.9 ± 0.3	65.0 ± 0.2	65.0 ± 0.2	**66.7 ± 0.1**
Cloud-S3	mIoU	61.7 ± 0.7	**63.0 ± 0.8**	60.9 ± 0.0	–	–	58.2 ± 0.1	62.0 ± 0.7
EuroSAT-S1	OA	81.7 ± 0.7	81.5 ± 0.9	75.4 ± 0.4	83.6 ± 0.1	83.9 ± 0.1	81.7 ± 0.1	**87.2 ± 0.1**
EuroSAT-S2	OA	97.5 ± 0.0	97.6 ± 0.1	92.5 ± 0.1	96.7 ± 0.0	97.0 ± 0.1	97.2 ± 0.1	**97.9 ± 0.1**
BigEarthNet-S1	mAP	71.5 ± 0.4	70.6 ± 0.4	63.8 ± 0.1	**78.7 ± 0.0**	70.8 ± 0.0	70.5 ± 0.0	77.9 ± 0.0
BigEarthNet-S2	mAP	79.5 ± 0.5	80.1 ± 0.1	71.6 ± 0.1	**83.6 ± 0.0**	76.4 ± 0.0	75.5 ± 0.0	79.0 ± 0.0
LC100Cls-S3	mAP	91.3 ± 0.3	91.4 ± 0.5	88.9 ± 0.1	–	–	89.5 ± 0.0	**93.3 ± 0.4**
DFC2020-S1	mIoU	49.9 ± 0.4	50.8 ± 0.5	45.4 ± 0.1	**52.8 ± 0.6**	**52.7 ± 0.1**	49.7 ± 0.1	52.4 ± 0.1
DFC2020-S2	mIoU	65.3 ± 0.6	66.2 ± 0.7	62.3 ± 0.0	64.1 ± 0.3	**66.5 ± 0.0**	61.8 ± 0.1	64.5 ± 0.1
LC100Seg-S3	mIoU	20.1 ± 0.4	19.3 ± 0.5	18.2 ± 0.1	–	–	16.5 ± 0.1	**24.1 ± 0.0**
Flood-S1	mIoU	78.0 ± 0.1	**78.3 ± 0.3**	75.1 ± 0.1	77.2 ± 0.1	77.4 ± 0.1	76.0 ± 0.1	77.7 ± 0.0
LCZ-S2	OA	**86.6 ± 0.7**	85.3 ± 0.8	77.4 ± 0.1	83.6 ± 0.2	84.1 ± 0.0	83.0 ± 0.3	84.4 ± 0.0
Biomass-S3	RMSE ↓	68.1 ± 0.3	68.3 ± 0.4	68.7 ± 0.5	–	–	74.1 ± 0.1	**66.3 ± 0.1**
AQ-NO2-S5P	RMSE ↓	3.4 ± 0.0	3.4 ± 0.0	3.4 ± 0.0	–	–	3.3 ± 0.0	**2.8 ± 0.0**
AQ-O3-S5P	RMSE ↓	1781.3 ± 29.8	1766.8 ± 22.1	1741.6 ± 11.5	–	–	1755.6 ± 19.8	**789.4 ± 2.6**

datasets for S3/S5P. The detailed characteristics are shown in Tab. 3. The curation process for each dataset and detailed benchmark analyses can be found in the appendix.

5.2. Benchmark results

We benchmark supervised training and frozen-encoder evaluation of several representative single-, dual-, and multimodal foundation models on Copernicus-Bench. For classification tasks, we perform linear probing with batch size 64 and the SGD optimizer for 50 epochs; for segmentation and regression tasks, we train a UPerNet [61] decoder on top of the frozen encoder with batch size 16 and the AdamW optimizer for 50 epochs; for change detection tasks, we separately encode the pre- and post-event images and calculate their feature map differences as input to a UPerNet decoder, hyperparameters following the segmentation design. For supervised baselines, we train the model from scratch for 80 epochs. We conduct a simple grid search for the learning rates and report test metrics under the best validation scores. All experiments are conducted on a single GPU with three runs to get the mean and standard deviation. The results are shown in Tab. 4. It can be seen that our Copernicus-FM is comparable to or better than state-of-the-art single- or multimodal foundation models on their appli-

Table 5. Linear regression results on 10-year mean and standard deviation (variability) of 6 climate parameters. We report three-run average RMSE scores with standard deviation (error bar). Best scores in **bold**.

Input Source	Temperature (°C)		Precipitation (m)		Surface press. (Pa)		Sea-level press. (Pa)		u wind (m/s)		v wind (m/s)	
	Avg	Std	Avg	Std	Avg	Std	Avg	Std	Avg	Std	Avg	Std
Coord. (raw)	8.36 ±0.00	3.06 ±0.00	0.79 ±0.00	0.50 ±0.00	7305.91 ±0.37	141.38 ±0.01	290.68 ±0.02	164.84 ±0.01	1.12 ±0.00	0.55 ±0.00	0.95 ±0.00	0.58 ±0.00
Coord. (FE)	3.99 ±0.00	2.09 ±0.00	0.61 ±0.00	0.42 ±0.00	6537.54 ±1.78	101.09 ±0.18	200.93 ±0.09	108.12 ±0.01	0.96 ±0.00	0.51 ±0.00	0.80 ±0.00	0.54 ±0.00
Embed.	2.66 ±0.01	1.72 ±0.00	0.47 ±0.00	0.33 ±0.00	**1627.21** ±30.13	97.53 ±0.71	197.17 ±0.62	117.57 ±0.86	0.94 ±0.00	0.43 ±0.00	0.78 ±0.00	**0.43** ±0.00
Embed. + coord.	**1.98** ±0.01	**1.35** ±0.00	**0.46** ±0.00	**0.32** ±0.00	1676.39 ±33.64	**78.53** ±0.08	**174.80** ±0.21	**90.37** ±0.15	**0.91** ±0.00	**0.42** ±0.00	**0.76** ±0.00	**0.43** ±0.00
Coord. (SatCLIP [33])	3.67 ±0.03	1.58 ±0.00	0.48 ±0.00	**0.32** ±0.00	3178.91 ±8.75	97.74 ±0.19	202.76 ±0.27	110.71 ±0.41	0.92 ±0.00	0.43 ±0.00	0.79 ±0.00	0.45 ±0.00

cable tasks, and largely improves downstream performances on S3 and S5P tasks. Encouragingly, our results outperform supervised training on the majority (11/15) of tasks, despite utilizing fewer trainable parameters and iteration steps. The results also verify the benefits of cross-modal pretraining on both surface and atmospheric applications.

6. Bridging EO & climate via grid embeddings

While Copernicus-FM has demonstrated its benefits for various EO tasks, one of its most exciting potentials lies in bridging EO with weather and climate analysis—an intersection that is still largely unexplored. Thanks to the grid-based dataset structure in Copernicus-Pretrain, we achieve a direct alignment between EO imagery and climate parameters from ERA5[29], one of the most widely used reanalysis datasets for weather and climate research. This allows Copernicus-FM-encoded grid embeddings to serve as semantically rich geographical representations that could potentially enhance climate modeling.

To evaluate this potential, we curate a simple set of climate tasks predicting the 10-year mean and standard deviation of 6 important climate parameters (2 m mean air temperature, annual total precipitation, surface pressure, mean sea-level pressure, and u/v component of 10-meter wind speed) using geocoordinates or geographical representations. We randomly sample 10K grids around the globe, split them into train/val/test subsets, and calculate the corresponding climate parameters from ERA5 as targets. As a baseline, the central coordinates of the grids are sent through a linear regression model for prediction. As shown in Tab. 5, the two baseline models with raw or Fourier-encoded coordinates as input verify the basic correlation between geography and climate. We then use Copernicus-FM-derived grid embeddings as input, obtained by simply averaging model-encoded images from various modalities. These embeddings provide a richer, high-level representation of each grid's environmental characteristics and consistently outperform coordinate-based baselines. Moreover, combining grid embeddings with coordinates yields the best overall performance, demonstrating the complementary nature of spatial context and EO-derived semantic features. Notably, even when compared against location encodings from a specialized location encoder trained on EO images [33], the grid embeddings still outperform, emphasizing the unique value of EO representations for climate studies. We extend a global embedding dataset at 0.25° based on these promising findings (see appendix for details). Looking ahead, we envision incorporating these grid embeddings into ML-based training for medium-range weather forecasting, expanding the set of static variables with EO-derived visual representations, or dynamic variables with representation time series.

7. Conclusion

This work presents a series of efforts towards next-generation EO foundation models, advancing existing approaches in data, model, and benchmark. We introduce **Copernicus-Pretrain**, extending EO pretraining datasets to all major Sentinel missions, encompassing both Earth's surface and its atmosphere. We propose **Copernicus-FM**, leveraging dynamic hypernetworks to process any spectral or non-spectral modality, enhancing adaptability across diverse EO sensors. We establish **Copernicus-Bench**, a comprehensive benchmark with hierarchical downstream tasks across various Sentinel modalities. Furthermore, we demonstrate the benefits of EO grid embeddings in a simple climate prediction task, highlighting their potential in bridging EO with weather and climate studies. This paves the way towards unified EO and climate foundation models, unlocking new possibilities for integrated Earth system understanding in future work.

As a limitation, the scope of this work is restricted to the Sentinel series and a temporal range of about 1 year. Future work will thus also explore efficient scaling for satellite expansion, as well as native model support for multimodal fusion and time series processing.

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